

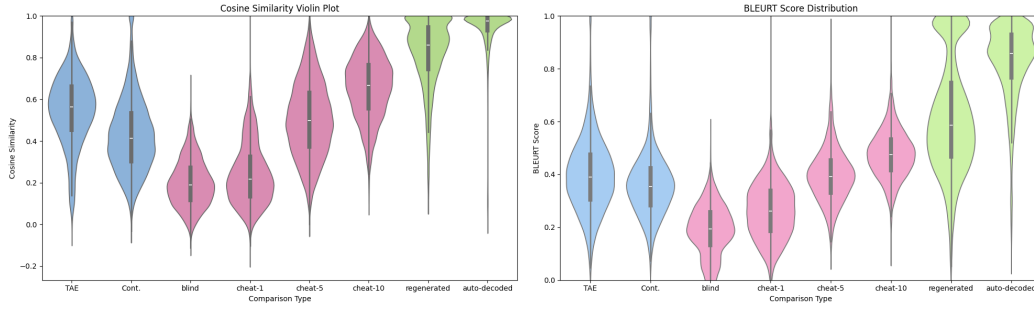


Figure 4: Violin plots showing the performance of TAE ParaScope and Continuation ParaScope against the baselines (0, 1, 5, and 10 cheat tokens) and ground truth (regenerated, auto-decoded) on Cosine sim (left) and BLEURT (right)

However, when looking at details match, we find an interesting difference. When it comes to achieving at least Minimal Depth Details (score ≥ 1 , Basic shared details without specifics), both TAE (57%) and Continuation (52%) ParaScopes perform similarly well to each other, and similar to a cheat-5 baseline (55%). However, when it comes to achieving at least Moderate Depth details (score ≥ 2 , Shared details with some supporting facts), Continuation ParaScope (15%) is significantly better than the TAE ParaScope (3%), which in turn falls just short of a 5-token baseline (17%). Even ground truth regenerated outputs often fell short, only 79% of which achieved a score ≥ 2 .

This is qualitatively in line with results from manually looking at examples (see Appendix I), where Continuation ParaScope can often either completely miss the task or can be quite accurate with the continuation.

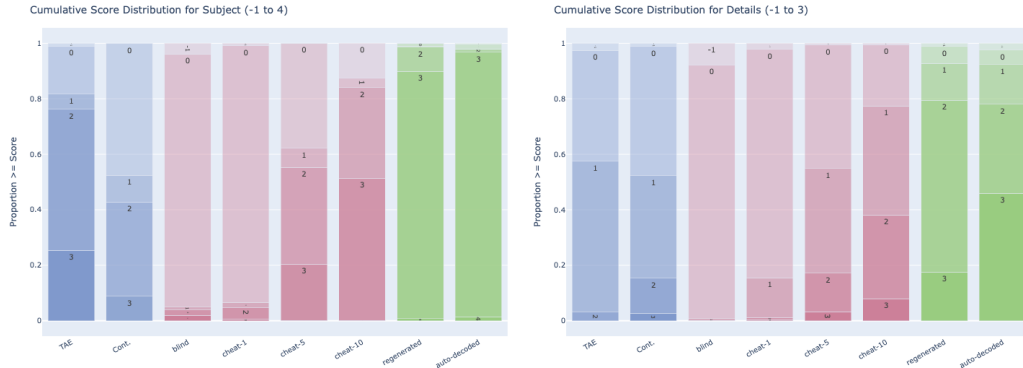


Figure 5: Cumulative bars showing the performance of TAE ParaScope and Continuation ParaScope against the baselines (0, 1, 5, and 10 cheat tokens) and ground truth (regenerated, auto-decoded). (left) shows subject match to original paragraph on a scale from -1 to 4, and (right) shows detail preservation on a scale from -1 to 3.

Importantly, both ParaScope methods show significant performance above random baselines, often being able to reconstruct key details about the upcoming text. TAE ParaScope seems to preserve general subject match well, and mostly gets approximate details correct, while Continuation ParaScope seems to work inconsistently but can sometimes restore more fine-grained details. These results give positive evidence for the Decodability Hypothesis, while falling short of ground-truth baselines.

5 OUTLINE-LEVEL RESIDUAL STREAM DECODERS

We extend the Decodability Hypothesis to outline-level representations. We take the method of SONAR ParaScopes to construct a TAE Outline RSD, which aims to map residual at the start of generation to a general outline of the text the model is likely to produce.

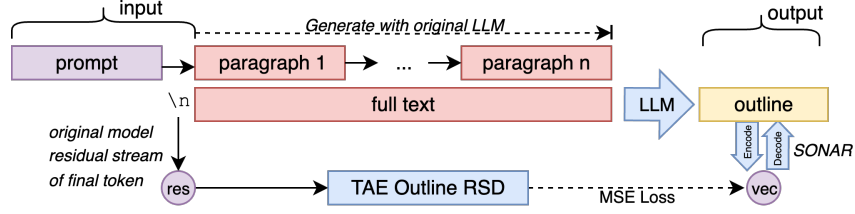


Figure 6: Diagram of experiments for TAE Outlines RSD. We create an outline of the generated dataset, and encode this outline with SONAR. We then train a linear map between the residual stream diffs of the model to the SONAR TAE embedding vector.

We construct a dataset of outlines for this experiment. We take the dataset of model generations we had from Section 3.1, and use Llama 3.2 70B to summarize the text into a bullet-point outline of key details. We then encode these outlines into a SONAR TAE vector to be used as the training target. As the training input, we use the model residuals at the newline token after the prompt.

The architecture for TAE Outline RSD is a linear map identical to that of TAE ParaScope (see Appendix G.1), but this time mapping the normalized residual stream diffs to SONAR TAE embeddings, which contain the outline of the upcoming document.

We then evaluate this with an LLM-as-a-Judge (GPT-4o-mini, as before) and compare it against ground-truth outlines generated by another model following the same process, Gemma 3 27B Instruct (Team et al., 2025). We focus mainly on Coverage of Key Points, Ordering/Flow, Subject Match, Entities Match, and Details Match. See the full rubric in Appendix E.

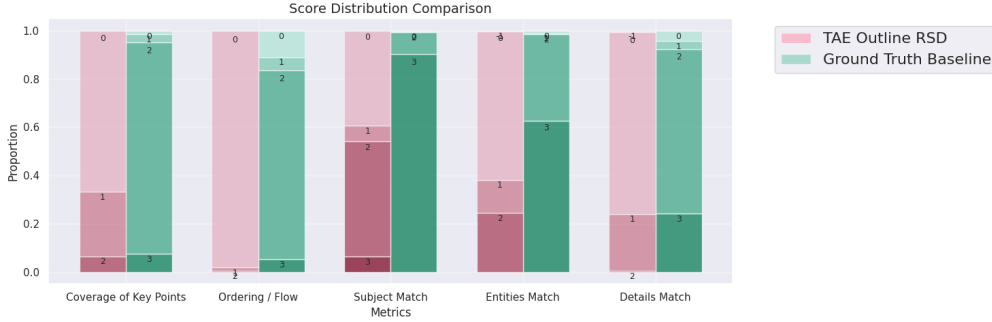


Figure 7: LLM-as-a-Judge comparison between outlines decoded by the TAE Outline RSD scored against the original Llama 3.2 70B outline. We compare against ground truth, which are the scores achieved by Gemma 3 27B with access to the full text & prompt.

We show the results in 6. We find that the TAE Outline RSD is able to decode some amount of information from the residuals stream, but that the performance is not as good as it was with the TAE ParaScope. We find that the subject match is generally high (61% at least minor match, 54% at least moderate match), but detail preservation is worse (24% minimal depth, i.e., very limited overlap), which is lower than for TAE ParaScope, which achieved minimal details 52% of the time.

Results on planning at the outline scale are relatively weak, aside from matching general subject matter. See example outputs in Appendix J. This gives some evidence for the Decodability Hypothesis, showing that the relatively small 3B-parameter model either does not plan at the outline scale or that such information is not linearly decodable. Experiments in the next section will provide some evidence for the former over the latter.

6 DEEPER ANALYSIS INTO INFORMATION AVAILABILITY

With this evidence for the Decoability Hypothesis, we try to apply the ParaScopes to understand when the model may be forming planning information.

6.1 LAYER-WISE INFORMATION ANALYSIS

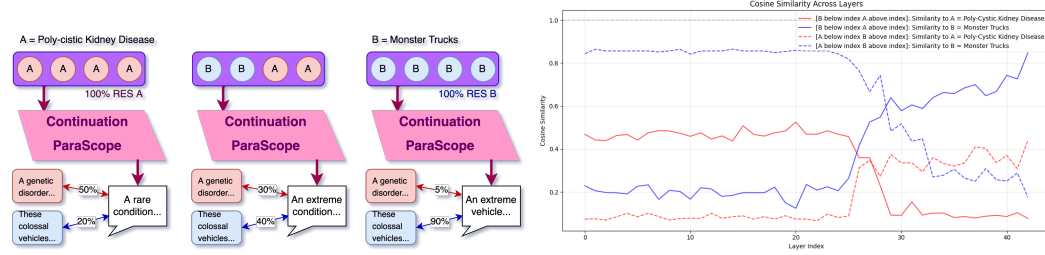


Figure 8: explanation of how we do "layer scrubbing" for layer-wise information analysis

We employ a simple causal scrubbing methodology (Chan et al., 2022) with Continuation ParaScope to determine which layers contribute most to paragraph planning. We take the activations of the model across all the layers for two examples, and attempt to interpolate between the two examples to observe when the model chooses to write about one versus the other. In particular, we take the following prompt:

Base Prompt: "Tell me about cleaning your house in 50 words, then tell me about [A or B] in 50 words. Structure it as 2 paragraphs, 1 paragraph each."

We design our experiments using a controlled prompt structure where the model first writes about cleaning a house, then transitions to one of two dramatically different topics: either polycystic kidney disease (Paragraph A) or monster trucks (Paragraph B). This stark contrast allows us to clearly distinguish which information is encoded at different layers.

For each layer K between 0 and N , we perform a layer scrubbing procedure that systematically isolates the contribution of different network depths. We first extract residual stream activations RES_A and RES_B from the paragraph transition token (" $\text{\textbackslash n\textbackslash n}$ ") for both test conditions.

We then create hybrid activations by combining layers $[1...K]$ from one condition with layers $[K + 1...N]$ from the other, generating outputs in both directions to capture the full effect of layer-wise information transfer. For each configuration, we generate 100 samples and compare the resulting outputs against reference texts using 'all-mpnet-base-v2' () embeddings.

To ensure our measurements reflect genuine information availability rather than limitations in our probing methodology, we generate 64 samples per condition and compare the best-matching outputs against references A and B.

This layer scrubbing analysis reveals a clear pattern of information processing. The early layers (0-25 out of 42 total) show minimal contribution to paragraph planning, with cosine similarity deltas below 0.05. The middle layers (25-35) demonstrate the strongest impact on future paragraph content, showing substantial cosine similarity deltas of 0.15 to 0.25. Finally, the output layer exhibits a distinct jump of approximately 0.1 in similarity delta, which we attribute to output embedding effects (directly influencing the first generation token) rather than planning computation. This suggests that layers 60%-80% into the model have the most planning-relevant activations.

6.2 TEMPORAL DYNAMICS: WHEN PLANNING HAPPENS

To investigate when paragraph information appears, we analyze token-wise dynamics around paragraph transitions and attempt to understand when the model starts encoding follow-up information.

We use Gemma-2-9B (Team et al., 2024) ¹ and perform Continuation ParaScope on each of the different tokens.

We perform a few steps: 1) Generate paired paragraphs with controlled pairs of topics. 2) Extract residual streams from tokens within ± 10 positions of "\n\n". 3) Apply Continuation ParaScope at each position. 4) Generate 64 possible completions per position. 5) Compare against original first or second paragraph using text embeddings.

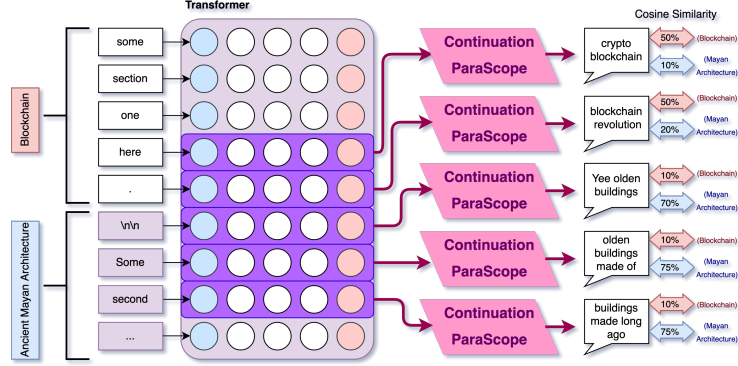


Figure 9: explanation of how we do token-wise analysis to see when the LLM prominently seems to be looking forward

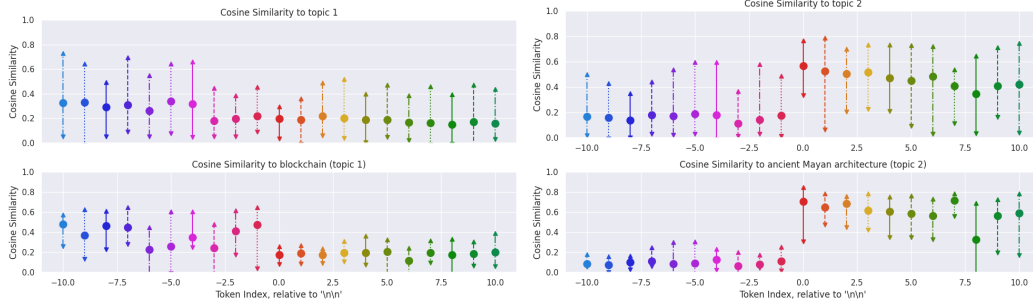


Figure 10: We get cosine similarity of Continuation ParaScope generations at tokens near the newline token. We compare similarity to paragraph 1 (left) and paragraph 2 (right), for a specific example (bottom) and averaged over 20 examples (top)

We first focus on an example where the initial topic is blockchain and the subsequent topic is ancient Mayan architecture, and in Table 6.2 compare the cosine similarities to both topics. We see that representation of the later topic only becomes significant at the newline token.

	Pre-transition	Transition token	Post-transition
Topic 1 (blockchain)	0.45–0.55	0.20–0.30	0.10–0.20
Topic 2 (architecture)	0.05–0.15	0.50–0.60	0.70–0.80

Table 1: Cosine similarity patterns across transition, before, at, and after ‘\n\n’, when using Continuation ParaScope.

Averaging across 20 different pairs of topics, we then look at similarity to Topic 2. We find pre-transition it is 0.12 (± 0.08), at transition it is 0.48 (± 0.15), and post-transition it is 0.65 (± 0.12). This gives moderate evidence that Gemma 2 9b is relatively myopic when it comes to forming plans in its activations, and mostly focuses on the specific immediate paragraph to write.

¹we wanted to do a more fine-grained test of tokens "." and "\n\n", and llama 3 combines these into one token